Dioxin congener patterns in commercial catfish from the United States and the indication of mineral clays as the potential source.

Since 1991 the US Department of Agriculture (USDA) has conducted annual surveys of pesticide residues in foods under the Agricultural Marketing Service's Pesticide Data Program (PDP). To assess chemical residues in domestically marketed catfish products, 1479 catfish samples were collected during the 2008-2010 PDPs. A subset of 202 samples was analysed for 17 toxic polychlorinated dibenzo-p-dioxins and furans (PCDD/Fs). The average pattern of the individual PCDD/F congener concentrations in the catfish was rather unique in that it had almost no measurable amounts of polychlorinated dibenzofurans (PCDFs), but all PCDDs were present. This pattern was more dominant in the domestically produced catfish products than in the imported products (China/Taiwan). Comparison of the pattern to known sources of PCDD/Fs showed strong similarities to the pattern of PCDD/Fs found in kaolin clays which have often been used as anti-caking agents in animal feeds. To investigate whether catfish feeds may be the source of the PCDD/Fs found in the catfish, archived catfish feed data from a US Food and Drug Administration (USFDA) database were examined. In 61 out of 112 feed samples, the PCDD concentrations were 50 times higher than the PCDF concentrations and resembled the pattern found in the catfish products and in clays mined in the south-eastern United States. Although the source of PCDD/Fs in domestically marketed catfish products cannot be definitively established, mined clay products used in feeds should be considered a likely source and, given the wide concentration range of PCDD/Fs that has been found in clays, a critical control point for PCDD/Fs entrance to the food supply.